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Stochastic Methods in Finance

Assignment – Spring Semester 2026

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Contents

1 Introduction

- Graph of appl (Andrea)
- Small introduction (Andrea)

2 Building the Binomial Tree

- graph of small tree only prices (Mathis) DONE
- explain how we build the tree (Juliette) DONE

We build the binomial tree as follows. The starting point is the closing price of Apple on April 28, 2026, denoted S_0 . At each period, the price either goes up (multiplied by u) or down (multiplied by d), where $\Delta t = T/n = 0.02$ years. Both factors are calibrated using the volatility σ estimated from Apple's daily log-returns over 2020–2026. The larger σ , the larger u and the smaller d .

Starting from S_0 , each node splits into two branches: an up-branch ($S \times u$) and a down-branch ($S \times d$). After 25 periods, the tree contains $2^{25} \approx 33$ million possible paths. At each node, we store the cumulative sum $C_t = \sum_{j=0}^t S_j$, needed to compute the arithmetic average $\bar{S}_n = C_n/(n+1)$.

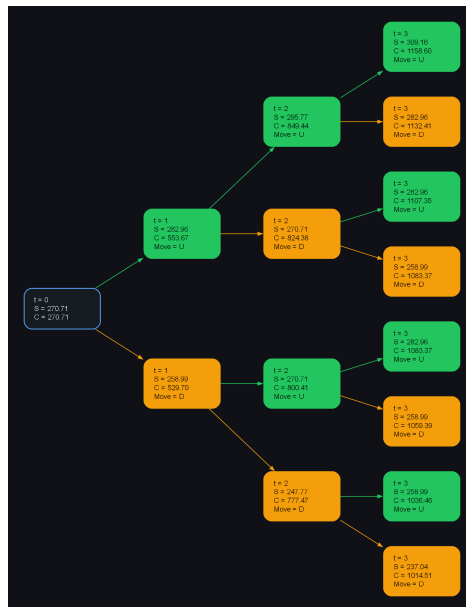


Figure 1: Binomial tree with 3 periods. Each node contains the stock price at that point in time.

- explain u and d (Mathis)
- explain how we compute the stock price at each node (Mathis)
- link number of nodes to number of periods $1+t$ (Juliette) DONE

At each period t , a recombining tree has $t+1$ distinct price nodes. At $t = 25$, this gives 26 distinct prices. However, since the Asian option payoff depends on the arithmetic average

$\bar{S}_n$, two paths reaching the same final price can yield different payoffs. We therefore track all $2^{25} \approx 33$ million paths separately, even though they converge to only 26 distinct prices at maturity.

3 Arithmetic Average at Terminal Nodes

- add formula for average (Mathis)
- add tree with average (Mathis)
- Explain how we got these results (Juliette) DONE

$$\bar{S}_n = \frac{1}{n+1} \sum_{j=0}^n S_j \quad (1)$$



Figure 2: Binomial tree with 3 periods. Each node contains the stock price and the average at that point in time.

To compute the arithmetic average at each terminal node, we use the cumulative sum C_t stored at each node throughout the tree. At maturity ($t = 25$), C_{25} contains the sum of all prices from S_0 to S_{25} . We then divide by $n+1 = 26$ to obtain the average $\bar{S}_n = C_{25}/26$.

- add distributions (2x2 graph): C (distribution + Paths), S-bar (distribution), S (distribution) (Mathis)
- Graphs descriptions (Andrea)
- calculate skewness and kurtosis of these distributions (Erwan)

4 Payoff at Terminal Nodes

- explain the payoff (Andrea)
- add formula (Andrea)
- Explain how we got these results (Juliette)

For each terminal node ($t = 25$), we already have the final price S_{25} and the arithmetic average $\bar{S}_n$ computed in the previous step. The option payoff is then $A_T = \max(S_{25} - \bar{S}_n, 0)$. If the final price exceeds the average, the option has value. Otherwise, the option expires worthless and the payoff is zero.

- Tree picture (With the payoff at the terminal nodes) (Mathis)
- Stats (done by Juliette)

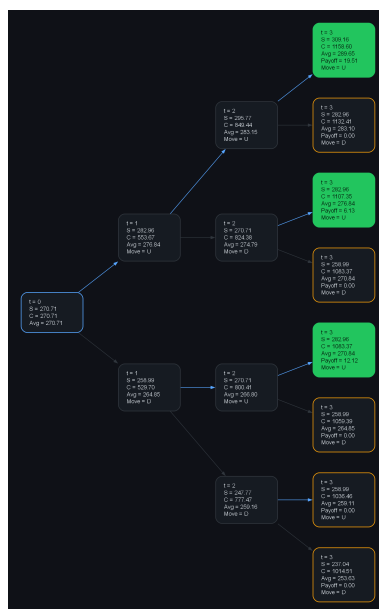


Figure 3: Extract of the first 3 nodes from the Binomial tree with 25 nodes. Here note that the payoff is only computed at the terminal nodes ($t=25$) and not at the intermediate nodes, the values are only for illustration and were not used for computing.

Table ?? reports descriptive statistics of the payoff $A_T = \max(S_n - \bar{S}_n, 0)$ computed at the terminal nodes of the binomial tree ($t = 25$).

Statistic	Value (\$)
Count	3.36×10^7
Mean	15.71
Std	24.71
Min	0.00
25%	0.00
Median	0.00
75%	24.65
Max	321.58

Table 1: Descriptive statistics of the payoff A_T at the terminal nodes ($t = 25$).

The distribution of A_T is highly right-skewed: the median is zero, meaning more than half of all terminal paths yield a worthless option, i.e. $S_n < \bar{S}_n$. Notably, the standard deviation (24.71) exceeds the mean (15.71), reflecting the high dispersion of payoffs. Most paths expire worthless while a small number of highly favourable paths drive the mean upward. The maximum payoff of 321.58 corresponds to the path where the stock price increases at every period.

- Payoff distribution (Terminal Nodes) (Mathis)
- Description and interpretation of the payoff distribution (Andrea)

5 Risk-Neutral Probabilities

- intro why we need q (Erwan)
- proof of q (Erwan)
- explain relation with vol and interest rate (Erwan)
- Matrix vol interest rate (Mathis)
- Interpretation of q matrix (Erwan)
- show result of q (Erwan)
- Distribution of the payoff under q vs p (Juliette)

Figure ?? displays the risk-neutral distribution of up moves over $n = 25$ steps, approximated by a normal distribution with mean $\mu_q = nq$ and variance $\sigma_q^2 = nq(1 - q)$.



Figure 4: Risk-Neutral Distribution of Up Moves ($n = 25$ steps, AAPL)

$q \approx 0.491$ is the risk-neutral probability derived from the no-arbitrage condition, depending on r and σ rather than historical returns. Unlike the physical probability p , which describes the actual likelihood of an up-move, q assigns relatively more weight to adverse states, reflecting investor risk aversion. The fact that $q < p$ is therefore expected.

Comparing skewness and kurtosis between the two is not straightforward, as differences in these moments capture not only beliefs about tail risks but also their pricing. With a skewness of 0.007 and excess kurtosis of -0.080 , the distribution of q is nearly symmetric and close to normal.

6 Backward Induction and Option Price

- explain backward induction depends on q and how we get the result (Mathis)
- graph of the tree with values (Mathis)
- Presentation of our result (Juliette) DONE

The arbitrage-free price of the Asian call option at $t = 0$ is obtained by applying backward induction over the entire binomial tree using the risk-neutral probability q :

$$V_0 \approx 14.05$$

This means that an investor should pay \$14.05 today to enter this contract. Note that V_0 is lower than the mean terminal payoff (15.71) because under the risk-neutral measure $q \approx 0.491$, the stock has slightly less chance of going up than under $p = 0.5$, leading to a lower expected payoff.

- show the price of the option being the same with different paths (Erwan)
- compare with MC (Andrea)

7 Robustness Check

- matrix with different q same as earlier before (Mathis)
- Compare price with MC 2std fixed strike (Andrea)
- Compare the different scenarios (Andrea)

8 Normal Approximation

- Values distribution (Mathis)
- Kurtosis and skewness of the distribution (Erwan)

9 Conclusion

- summary of the results (Andrea)
- limitations of the model (Andrea)
 - comparaison with MC (Andrea)
- possible extensions (Juliette) DONE
 - more iterations, more periods... (Juliette)

Our model relies on several simplifying assumptions that open the door to possible extensions. First, increasing the number of periods n would yield a more accurate price. With $n = 25$, the result is an approximation and the larger n , the more V_0 converges to the true value. Second, extending the maturity beyond 6 months would allow us to analyse how the option price evolves over time. Finally, using a stochastic volatility instead of a constant σ would better reflect Apple's real behaviour. However, this extension is computationally demanding in our case, as the path-dependency of the Asian option combined with a time-varying volatility would cause the tree to grow exponentially.

- final remarks (Erwan)